

READ THE COMMAND WORD FIRST — IT DECIDES THE SHAPE		
THE STEM SAYS	WHAT EARNS THE MARKS	FROM THE PAPERS
“State...” or “Identify...”, 1 mark	the name only. Extra words cannot add marks and cost time	“State the model of DNA replication” (1) → semi-conservative. One word
a stem that names a list	marks = number of items listed. One point each, no verdict anywhere	“the following structures: Nucleus, Golgi apparatus, Secretory vesicles” (3) → 3 items, 3 marks
“Explain why...”, “Explain how...”, “Explain the role of...”	no verdict exists. Write one linked causal step per mark	“Explain how this affects the nervous system” (3) → 3 steps, no decision
a decision plus “explain your answer” or “justify your answer”	the verdict in the first three words, then separate reasons until there are as many as the mark value	the most common shape, 15 of 32 marked parts
“Justify...”	identical to “explain your answer”, but every use on the papers is worth 3 or more	the word is a reliable signal that the part is expensive
“Predict and explain...”	- what is normally true - what the change breaks - the consequence that follows	naming the normal state is a scoring beat , not preamble

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THE STEM SAYS	WHAT EARNS THE MARKS	FROM THE PAPERS
“Deduce...” with a table	the verdict, and the figure or table must be quoted	the only command word grammatically bound to data
“Do you agree?”, “Would X...?”	yes or no in the first three words , then the reason	a hedge scores nothing . A wrong verdict with a sound chain still collects the chain
“Show your workings clearly”	the arithmetic line earns independently of the answer	“10 cells, 5 rounds” (2) → write $10 \times 2^5 = 320$
never used on this paper	describe, suggest, compare	describe appears twice, both inside multiple-choice stems
HOW THE MARKS ACTUALLY DIVIDE		
32 marked parts, 78 marks, across five papers. The other six papers print no mark values at all.		
The 3-mark part is the centre of gravity — eleven parts, 33 marks, 42% of the total. Nothing at part level is worth more than 5. A one-sentence justification on a 3-marker caps at about a third of what is available.		
COMMITTING TO A VERDICT The stem forces a closed choice: leading or lagging, Q or R, enhanced or impaired, monocistronic or polycistronic . Use one of those exact printed words. Paraphrasing a forced choice throws the verdict mark away.		
82% of the marks are figure-led — 64 of 78, and three of the five marked papers are entirely so. That is the part no textbook can supply.		
WHERE MARKS ARE LOST, IN ORDER OF EXPOSURE		
THE MISTAKE	WHAT IT COSTS	THE FIX
writing a verdict where none was asked	23 marks , 29% of the paper	on “explain why/how/the role of” there is nothing to decide. Spend every sentence on a causal step

WHERE MARKS ARE LOST, IN ORDER OF EXPOSURE		
THE MISTAKE	WHAT IT COSTS	THE FIX
answering at the wrong level of organisation	total loss when it happens	“State the consequence at the organ level ” is not answered by a correct cell-level sentence
answering the figure from a textbook	up to 9 marks on one question	the lac operon figure is a MUTANT . The canonical diagram gives the opposite answer
not quoting the figure or table	64 marks are figure-led	an answer that would read the same with the figure removed has not answered the question
hedging on a forced choice	all 15 decision parts	there is no partial-credit shape for “it depends”
one reason where two or three are wanted	the whole 3-mark band	separate sentences. One long sentence with three clauses reads as one point
over-writing the cheap marks	24 marks sit in 1-mark recall parts	a 1-mark “state” wants one word. Minutes spent decorating it are minutes off the 5-marker
THE SENTENCE SHAPES THAT RECUR		
DECISION PLUS JUSTIFICATION, 60% OF THE PAPER The verdict. Because reason one . Because reason two . Because reason three .		
CAUSAL CHAIN, ONE ARROW PER MARK Normally X does Y . Without X , Y can no longer happen. So consequence , and therefore further consequence .		
ANY FIGURE-LED QUESTION From Figure 1 , the label or value seen . This means the mechanism . Therefore the conclusion .		
WHEN THE STEM NAMES A LEVEL At the organ level, the organ-level outcome . This is because at the cell level the cell-level cause , and since every cell in the gland is affected, the organ fails .		
The one-line test: if the sentence contains no because and no so that , it is not finished.		

WHICH ORGANELLE — FOLLOW THE ROAD		
SIGNATURE	METHOD	EXAMPLE
gene, DNA, mRNA made	NUCLEUS	stores DNA as chromatin; RNA is synthesised here and leaves through the nuclear pores
a protein that will be secreted	ROUGH ER	ribosomes on the RER make secreted and membrane proteins
a protein that stays in the cytosol	free ribosome	the discriminator is destination, not the protein
lipid, cholesterol, phospho-lipid	smooth ER	no ribosomes
modified, sorted, packaged	GOLGI	receives from the RER in transport vesicles, forms secretory vesicles
delivered and released	SECRETORY VESICLE	fuses with the plasma membrane, exocytosis
digestive enzymes, breaking macro-molecules down	lysosome	
oxidative reactions, fatty acids, H ₂ O ₂ and catalase	peroxisome	also the initial rounds for fatty acids of 22 carbons or more
plant against animal	cell wall, chloroplast, large vacuole = plant	the deck lists centrosome absent in plant cells; that version wins

ORGANISM	the whole being
ORGAN SYSTEM	e.g. digestive
ORGAN	e.g. stomach
TISSUE	e.g. muscle
CELL	the unit of life
ORGANELLE	e.g. mitochondrion
MOLECULE	e.g. DNA
ATOM	smallest unit

Answer at the level the question names. A correct cell-level answer to an organ-level question scores nothing.

WORKED — THE SECRETORY ROUTE, CARRYING ONE NAMED PRODUCT

Casein production in a mammary epithelial cell 3

“State and explain the role of the following structures for the production and secretion of caseins. (3 marks) Nucleus / Golgi apparatus / Secretory vesicles”

1. The stem **names three items for three marks**, so this is one point each and there is no verdict to give
2. **Nucleus** — holds the casein gene and is where it is transcribed; the mRNA leaves through the nuclear pores to be translated
3. **Golgi** — receives casein from the rough ER in transport vesicles, then modifies, sorts and packages it into secretory vesicles
4. **Secretory vesicles** — carry the packaged casein to the plasma membrane and fuse with it, releasing casein into the lumen by exocytosis
5. **A bare function statement scores about half.** The mark is in linking each organelle to **casein** and to the **next step**

Three structures, three marks, each with its role AND what it hands on

WORKED — ANSWERING AT THE LEVEL THE STEM NAMES

Consequence of losing the rough ER 3

“State the consequence **at the organ level** with regards to milk production

if rough endoplasmic reticulum shown in Figure 1 is absent. Justify your answer. (3 marks)”

1. Read the noun: the question asks about the **ORGAN**. A correct cell-level answer scores nothing here
2. Organ-level statement: **the mammary gland produces little or no milk**, or milk lacking casein — the gland fails to secrete milk protein into the lumen
3. Justify, synthesis: casein is a **secreted** protein, so it must be made on ribosomes bound to the rough ER. Without RER it cannot be synthesised
4. Justify, the rest collapses: with nothing leaving the RER, the Golgi has nothing to package and no secretory vesicles form
5. Scale it: **every** epithelial cell lining the gland is affected identically, so the failure runs from cell to tissue to organ

The first sentence must be about the gland, not the cell

MODEL WORDING — CELLS

ORGANELLE ROLE
The Golgi apparatus modifies, sorts and packages the casein received from the rough ER , so that casein can be enclosed in secretory vesicles for release from the cell.
ABSENCE CONSEQUENCE
Without the rough ER , translation of secreted proteins cannot occur, so casein is not made or delivered to the Golgi ; therefore at the organ level , the gland cannot secrete milk protein.
Quotable definition, idiosyncratic to this module: cytoplasm = cytosol + organelles – nucleus.

REPLICATION — WHO DOES WHAT		
SIGNATURE	METHOD	EXAMPLE
unwinds, opens the fork	HELICASE	breaks the hydrogen bonds between complementary bases, separating the two strands
relieves the strain ahead of the fork	topoisomerase	breaks, swivels and rejoins; prevents supercoiling
lays down the RNA primer	PRIMASE	supplies the free 3'-OH that polymerase needs to start
builds the new strand	DNA polymerase III	only ever 5' to 3'
removes the primer, fills the gap	DNA polymerase I	
seals the nick between fragments	LIGASE	forms the phosphodiester bond
“which strand”, leading or lagging	1. count the primers 2. one primer, continuous, towards the fork → leading 3. many primers, Okazaki fragments → lagging	the tell is the primer count, not the direction alone
A:T against G:C	G:C is stronger	3 hydrogen bonds against 2. More bonds, more energy to separate
purine or pyrimidine	Pur As Gold: A and G, two rings CUT the Py: C, U, T, one ring	

THE FORK

why there are two kinds

synthesis can only run 5' to 3', so the strand running the other way must be built backwards in pieces

the pieces

Okazaki fragments, joined afterwards by LIGASE

semi-conservative

each daughter molecule keeps ONE parent strand and one new one

Helicase UNWINDS. Polymerase BUILDS. Ligase JOINS. A named enzyme with its named job scores; a description of the process without a name usually does not.

WORKED — IDENTIFY THE STRAND AND JUSTIFY

Where the RNA nucleotides are 4

“RNA nucleotides were observed to be more abundantly base paired to one strand of the DNA during replication. Identity [sic] the DNA strand (leading or lagging strand) and justify your answer. (4 marks)”

1. Use one of the two printed words. The verdict is **the lagging strand**

2. The RNA nucleotides are **RNA primers**, laid by primase to give DNA polymerase III a free 3'-OH to extend from

3. DNA is only ever built **5' to 3'**, so on this template synthesis runs away from the fork and must be **discontinuous**, as Okazaki fragments

4. **Each fragment needs its own primer**, so many primers lie along the lagging strand

5. The leading strand is continuous and needs only **one** primer at the origin — which is exactly the observation in the stem

Lagging strand, because one primer per Okazaki fragment against one in total

WORKED — STATE AND JUSTIFY A MITOSIS STAGE

Reading the micrograph 4

“State and justify the stage of mitosis shown in the cell in Figure 1. (4 marks)”

1. The stage is **anaphase**

2. The sister chromatids have **separated** and are no longer aligned at the metaphase plate; each is now a separate chromosome

3. They form **two groups moving towards opposite poles**, with a clear gap and spindle fibres visible across it — which rules out metaphase

4. The **nuclear envelope has not reformed** and the chromosomes have not decondensed — which rules out telophase

5. **Ruling out the neighbouring stage is usually its own mark** on a 4-mark justify

Anaphase, justified by what is visible rather than by a definition

READING A MITOSIS FIGURE IN THREE SECONDS

WHAT IS VISIBLE	STAGE	THE CONFIRMING DETAIL
condensed but scattered, envelope gone	prophase	spindle forming
one line at the equator	metaphase	a single plane
two groups with a gap	anaphase	spindle visible between them
two clusters at the poles, decondensing	telophase	envelopes reforming
chromosome count after division	unchanged	10 in, 10 out. Replication doubles the DNA, not the chromosome number
cells after n rounds	starting number × 2ⁿ	10 × 2 ⁵ = 320. Write the line

THE LAC OPERON — ONE QUESTION SETTLES IT		
SIGNATURE	METHOD	EXAMPLE
the only question worth asking	can the repressor physically sit on the operator?	yes → operon OFF no → operon ON
the parts	Pi promoter for lacI lacI makes the repressor Plac promoter for the structural genes O operator	structural genes: lacZ beta-galactosidase, lacY permease, lacA transacetylase
normal, no lactose	repressor made, binds the operator	transcription OFF
normal, lactose present	lactose binds the repressor and changes its shape	repressor cannot bind → ON
Pi deleted (the examined figure)	no repressor is ever made	operator always free → ON, constitutively
lacI deleted	no repressor protein	ON
operator mutated	repressor cannot recognise it	ON
“mono-cistronic or poly-cistronic”	count structural genes under one promoter	three genes, one promoter, one mRNA → polycistronic

WORKED — THE MUTANT OPERON

Transcription in the absence of lactose 5

“Figure 1 shows the lac operon in a mutant E. coli. In the absence of lactose, would transcription take place at the lac operon in this mutant E. coli? Explain your answer. (5 marks)”

1. Compare the printed figure with the normal map first. The figure

reads lacI, Plac, O, lacZ, lacY, lacA — Pi is missing

2. The verdict: **yes, transcription would take place**, even with no lactose

3. Without Pi, RNA polymerase cannot bind upstream of **lacI**, so lacI is not transcribed and **no repressor protein is made**

4. With no repressor in the cell, nothing binds the **operator**, so it stays free

5. RNA polymerase can therefore bind **Plac** and move along without steric hindrance, so lacZ, lacY and lacA are transcribed **constitutively**

Yes — the operon is permanently ON. Four of the five marks are in the chain, so write the chain even if the verdict is uncertain

EPIGENETICS — THE TRAP IS THAT NOTHING CHANGES

One examined part gives 4 marks for four one-word answers and asks for no explanation at all. When histone acetylation of a gene increases: gene sequence **NO** RNA sequence **NO** protein sequence **NO** amount of protein **YES**

Anyone reasoning “acetylation affects expression, so something must change” gets three of the four wrong. Epigenetic means a change in gene activity **without** a change in the DNA sequence.

IF THE QUESTION GIVES ROOM

Acetylation neutralises the **positive** charge on the histone, loosening its grip on the negatively charged DNA. Chromatin opens from **heterochromatin to euchromatin**, so transcription machinery can reach the gene and **more** mRNA is made.

Methylation against acetylation: methylation acts on the **DNA** (cytosine) and blocks **recognition**; acetylation acts on the **histone** (lysine) and changes **access**. Methylation is long-term silencing; acetylation is rapidly reversible.

WORKED — DEDUCE FROM A TABLE

Memory genes in mice X 4

“Low histone acetylation levels in the memory genes were found to be necessary for memory function. Deduce if mice X has enhanced OR impaired memory function, compared to the normal mice. Explain your answer. (4 marks)”

Table 1: gene A + to +++, gene B + to +++, gene C + to +++

1. Use a printed word. The verdict is **impaired**

2. Quote the table. Expression of all three genes is **higher** in mice X: A from + to +++, B from + to +++, C from + to +++

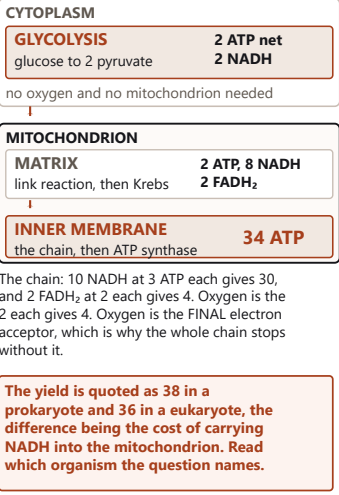
3. Higher expression implies **higher acetylation**, because acetyl groups neutralise the histone charge, open the chromatin and let transcription machinery in

4. The stem states that **low** acetylation is necessary for memory. Quote that rule back

5. Mice X has high acetylation, so the condition is not met

Impaired. The reasoning runs backwards: expression is observed, acetylation is inferred

GLYCOLYSIS, FERMENTATION AND THE KREBS CYCLE		
SIGNATURE	METHOD	EXAMPLE
glycolysis, per glucose	2 ATP in, 4 out, net 2 2 NADH 2 pyruvate	in the cytoplasm , and it runs with or without oxygen
pyruvate to acetyl-CoA, per glucose	2 NADH and 2 CO₂	in the mitochondrial matrix
Krebs, per turn	2 CO₂, 3 NADH, 1 FADH₂, 1 GTP	the deck writes GTP as equivalent to ATP; say it that way
Krebs, per glucose (two turns)	4 CO₂, 6 NADH, 2 FADH₂, 2 ATP	one glucose gives two pyruvate, so the cycle turns twice
carbon counts round the cycle	citrate 6C , alpha-ke-toglutarate 5C , then 4C for succi-nate, fu-marate, malate, oxaloacetate	two decarboxyla-tions, two carbons lost
where ATP actually comes from	glycolysis, Krebs, and the chain. Fermentati on makes NONE	fermentation only regenerates NAD ⁺
lactate or ethanol	CO ₂ released → ethanol , yeast, two steps no CO ₂ → lactate , muscle, one step	ethanol route: pyruvate decar-boxylase, then al-cohol dehydrogenase
“which oc-curs only under anaerobic conditions”	fermenta-tion , not glycolysis	glycolysis runs under both , so it is never the answer



WORKED — WHY FERMENTATION EXISTS AT ALL

An enzyme X deficiency in muscle

“If a person has a deficiency in enzyme X in his muscle cells, predict and explain what will happen to ATP production.”

- Enzyme X catalyses the step where **pyruvate is reduced to lactate** and **NADH is oxidised back to NAD⁺**
- Without it, NADH cannot be re-oxidised, so the cytoplasmic supply of **NAD⁺ is exhausted**
- Glycolysis **requires NAD⁺** as its oxidising agent, so once none is left, **glycolysis halts**
- Under anaerobic conditions ATP production therefore falls to **zero** — not even the net 2 ATP per glucose survives
- Consequence: rapid fatigue and exercise intolerance

Fermentation makes no ATP itself; it exists solely to regenerate NAD⁺ so that glycolysis can keep going

WHICH FUEL, AT WHICH HOUR

GLYCOGENOLYSIS	0 to 24 hours
glucose from liver glycogen	
GLUCONEOGENESIS	1 to 3 days
glucose built from amino acids and glycerol	
KETOGENESIS	beyond 3 days
ketone bodies, which cross to the brain	
carbohydrate	1 748 kcal about 1 day
mobilisable protein	24 440 kcal about 15 days
triglyceride	135 945 kcal about 85 days
at a resting requirement of 1 600 kcal per day	
Fat cannot be turned into glucose, which is why protein must be broken down in the middle phase. Ketones spare that protein.	

MODEL WORDING — RESPIRATION

MEMORISE THIS ONE VERBATIM

Fermentation itself produces **no ATP**. Its role is to oxidise **NADH** back to **NAD⁺**, so that glycolysis can continue to produce a net **2 ATP** per glucose in the absence of oxygen.

MISSING ENZYME IN A PATHWAY

Without **the dehydrogenase enzymes, the oxidation steps** cannot proceed, so **NADH and FADH₂** are not produced and **oxaloacetate** is not regenerated. The cycle stalls, and because no carriers reach the chain, oxidative phosphorylation stops too.

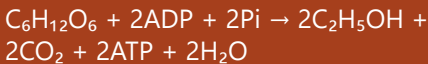
Always give both halves of a blocked pathway: what builds up **and** what runs out. Most answers give only one.

WORKED — A NET EQUATION WITH EVERYTHING SHOWN

Yeast, glucose to ethanol

“Write down the overall chemical equation for one molecule of glucose undergoing the metabolic processes shown in Figure 2. Show all reactants and products in their correct amounts.”

- Glycolysis: C₆H₁₂O₆ + 2ADP + 2NAD⁺ + 2Pi → 2 pyruvate + 2ATP + 2NADH + 2H⁺ + 2H₂O
- Decarboxylation: 2 pyruvate → 2 acetaldehyde + **2CO₂**
- Reduction: 2 acetaldehyde + 2NADH + 2H⁺ → 2 ethanol + 2NAD⁺
- NAD⁺ and NADH cancel** — they are regenerated, not consumed, so they must NOT appear in the net equation
- The stem says **all** reactants and products, so ADP, Pi, ATP and H₂O must all be written



THE ATP LEDGER — THE TABLE THE DECK LEAVES BLANK		
STAGE	CARRIER	ATP AFTER THE CHAIN
glycolysis	2 ATP direct	2
glycolysis	2 NADH	6
pyruvate to acetyl-CoA	2 NADH	6
Krebs	2 ATP direct (as GTP)	2
Krebs	6 NADH	18
Krebs	2 FADH ₂	4
TOTAL, prokaryote		38
TOTAL, eukaryote	less the shuttle cost	36
the reason for the difference	2 ATP are spent carrying glycolytic NADH into the mitochondrion	no mitochondrion, no cost

1 NADH — 3 ATP

1 FADH₂ — 2 ATP

1 acetyl-CoA — 12 ATP

fatty acid activation — -2 ATP

12 = 1 + 2 + 9, one turn of the Krebs cycle

PER GLUCOSE
prokaryote 38 eukaryote 36
the 2 difference is the shuttle cost

A textbook or a search gives 2.5, 1.5 and 30 to 32. That is a DIFFERENT CONVENTION, not a correction. Use the numbers above.

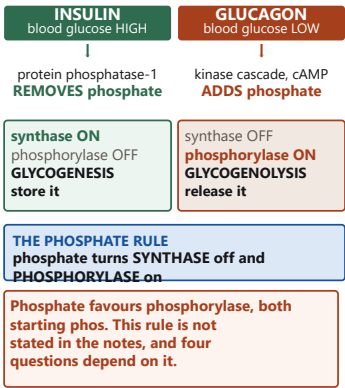
THE ELECTRON TRANSPORT CHAIN		
SIGNATURE	METHOD	EXAMPLE
what carries the electrons in	NADH and FADH₂	NADH enters at complex I, FADH ₂ at complex II
where it sits	inner mitochondrial membrane	in a prokaryote, the plasma membrane
what oxygen does	the final electron acceptor	combines with protons and electrons to make water. No oxygen, no chain
how ATP is actually made	protons return through ATP synthase	they cannot cross the inner membrane any other way
uncoupler, UCP	carries protons back, bypassing ATP synthase	ATP DOWN, electron flow UP, energy lost as heat
chain inhibitor	blocks a complex	electron flow down, ATP down
ATP synthase inhibitor	blocks the return path	ATP down, gradient builds, flow slows

WORKED — AN UNCOUPLER	
What UCP does to the chain	
“Uncouplers (shown as UCP in Figure 3) are hydrophobic molecules that are soluble in the bilipid membrane. They bind protons and transfer them across the bilipid membrane. How does this affect electron transport and ATP synthesis?”	
- The uncoupler carries protons from the intermembrane space back into the matrix without passing through ATP synthase - This dissipates the proton gradient, reducing the proton-motive force ATP synthesis falls, because fewer protons flow through ATP synthase Electron transport CONTINUES, and speeds UP — with a smaller gradient opposing them, the complexes pump more easily, so electron flow and oxygen consumption rise - The energy is released as heat instead of being captured as ATP	
Electron transport has been uncoupled from ATP synthesis. Step 4 is the discriminating mark and the one most answers get backwards	

GLYCOGEN		
SIGNATURE	METHOD	EXAMPLE
the bonds	alpha-1,4 along the backbone alpha-1,6 at branches	10 ³ to 10 ⁶ glucose units
why branched	many non-reducing ends exposed at once	enzymes attack many sites simultaneously, so mobilisation is fast
building it	glycogen synthase	glycogenesis
breaking it	glycogen phosphorylase	releases glucose-1-phosphate, NEVER free glucose
liver against muscle	does the tissue have glucose-6-phosphatase?	liver yes → free glucose into the blood, serves the whole body muscle no → G6P trapped, serves that muscle only

GLYCOGEN		
SIGNATURE	METHOD	EXAMPLE
which storage disorder	high glycogen + low blood glucose → breakdown blocked low glycogen + low blood glucose → synthesis blocked	enlarged liver means glycogen is accumulating

THE PHOSPHATE RULE, AND THE FIVE WORDS ONE LETTER APART		
THE WORD	THE SINGLE TELL	TRIGGER IN THE STEM
glycogenesis	glycogen is the product	insulin, fed state, synthase
glycogenolysis	glycogen is the sub-strate	it must still be present for this to run
gluconeogenesis	glycogen is absent entirely	substrates are lactate, glycerol, amino acids. Neo means new
lipolysis	the sub-strate is triglyceride	products are fatty acids and glycerol. No glucose is made
ketogenesis	the sub-strate is acetyl-CoA	the trigger word is always accumulates
THE PHOSPHATE RULE	phosphate turns SYNTHASE off and PHOSPHORYLASE on	phosphate favours phosphorylase, both starting phos. Not stated anywhere in the notes , and four questions depend on it
insulin	activates protein phosphatase-1 , which removes phosphate	synthase ON, phosphorylase OFF → storage
glucagon	activates the kinase cascade, which adds phosphate	synthase OFF, phosphorylase ON → release



FAT — WHICH DIRECTION, WHICH COMPARTMENT		
SIGNATURE	METHOD	EXAMPLE
the first question, always	is the word building or breaking?	synthesis, lipogenesis, malonyl-CoA, NADPH → cytosol oxidation, catabolism, carnitine → mitochondrial matrix
very long chains, 22 carbons or more	peroxisome, initial rounds only	uses acyl-CoA oxidase and makes H ₂ O ₂ , so catalase is needed
getting fat INTO the mitochondrion	carnitine shuttle	fatty acyl-CoA cannot cross the inner membrane by itself
per round of beta-oxidation	1 FADH ₂ , 1 NADH, 1 acetyl-CoA, chain 2 carbons shorter	
how many rounds	rounds = (n / 2) – 1, acetyl-CoA = n / 2	the last round splits a 4-carbon chain into two acetyl-CoA. This rule is not in the notes
the activation cost	–2 ATP	ATP goes to AMP + PPi, so two high-energy bonds are spent
acetyl-CoA to malonyl-CoA	1 ATP, by acetyl-CoA carboxylase	3 carbons in the malonyl group, but only 2 reach the chain – the third leaves again as CO ₂
why fat needs oxygen	beta-oxidation feeds NADH and FADH ₂ to the chain	sprinting is anaerobic and uses carbohydrate; endurance is aerobic and uses fat

A

D

B

C

PER ROUND
1 FADH₂
1 NADH
1 acetyl-CoA

A dehydrogenase makes FADH₂
B hydratase adds water
C dehydrogenase makes NADH
D transferase cuts off acetyl-CoA

For a fatty acid of n carbons:
rounds = (n / 2) – 1
acetyl-CoA = n / 2

The last round splits a 4-carbon chain into TWO acetyl-CoA, which is why the rounds are one fewer. This rule is not stated anywhere in the notes.

WORKED — COMPLETE OXIDATION OF STEARIC ACID

An 18-carbon fatty acid, end to end

“One molecule of stearic acid (18-carbon fatty acid) undergoes complete oxidation through Pathway X. List all the products in their correct amounts, then calculate the total number of ATP molecules produced.”

1. Rounds = (18 ÷ 2) – 1 = **8**. Acetyl-CoA = 18 ÷ 2 = **9**

2. Per round gives 1 each, so **8 NADH** and **8 FADH₂**

3. FADH₂: 8 × 2 = **16 ATP**

4. NADH: 8 × 3 = **24 ATP**

5. Acetyl-CoA: 9 × 12 = **108 ATP**, where 12 = 1 + 2 + 9 for one Krebs turn

6. Gross = 16 + 24 + 108 = **148**. Then subtract **2** for activation

7. Write the activation line explicitly — some sources count it as 1, and the working protects the method marks either way

Products: 9 acetyl-CoA, 8 NADH, 8 FADH₂. Total = **146 ATP**

WORKED — MALONYL-COA, AND THE CARBON THAT LEAVES AGAIN

Why the cell spends an ATP before it builds

“How many carbon atoms are added to the fatty acid chain from one molecule of malonyl-CoA? Explain.”

1. The acetyl group has **2** carbons; the malonyl group has **3**

2. **Two** carbons are added to the growing chain

3. The third carbon is the one added as CO₂ by acetyl-CoA carboxylase, and it is **released again as CO₂** during the condensation step

4. The carboxylation and then decarboxylation is what makes the condensation **energetically favourable**

5. That is why an ATP is spent making malonyl-CoA rather than condensing acetyl-CoA directly

Two carbons. Chains always grow in 2-carbon units

THE PAIRED TRAP, AND THE MODEL SENTENCES

THE CHOICE	THE TELL	NOTE
synthesis or breakdown compartment	these are set as consecutive questions, each with the other answer as a distractor	synthesis → cytosol breakdown → mitochondria
storage form in adipose tissue	triglycerides	
main membrane lipid	phospholipids	hydrophobic, so insoluble in water
why lipids differ from other macromolecules	they are nonpolar	
energy density	fat 9 kcal/g against carbohydrate 4	and glycogen binds about 2 g of water per gram; fat binds none
None		

MODEL WORDING — LIPIDS

COMPARTMENT ANSWER

Fatty acid synthesis occurs in the **cytosol** of **liver and adipose** cells, whereas beta-oxidation occurs in the **mitochondrial matrix**. The carnitine shuttle is required because **fatty acyl-CoA** cannot cross the inner mitochondrial membrane by itself.

STATING A BETA-OXIDATION YIELD

One molecule of a **18**-carbon fatty acid undergoes **8** rounds, producing **9** acetyl-CoA, **8** NADH and **8** FADH₂.

Peroxisomes also perform beta-oxidation, but only the initial rounds on very long chains of **22 carbons or more**. If a question adds “very long chain”, the answer flips from mitochondrion to peroxisome.

TRANSCRIPTION AND TRANSLATION		
SIGNATURE	METHOD	EXAMPLE
codon or anticodon	letters drawn ON the tRNA are the ANTICODON N	never look an anticodon up in the table. Read the mRNA underneath, or complement it
which strand is the template	if it is written 5' to 3', REVERSE it first	transcribing as printed gives a completely different, wrong peptide
where translation starts	the first AUG	everything before it is untranslated
where it stops	UAA, UAG, UGA	a stop codon codes for no amino acid
the three RNAs	mRNA carries the message tRNA brings the amino acid rRNA builds the ribosome and catalyses the bond	a 3-mark question wants all three, each with its role
the bond formed	peptide bond	covalent, and heat-resistant
why 3 letters	$4^3 = 64$ combinations for 20 amino acids	4^2 and 4^2 are too few
prokaryote against eukaryote	eukaryote: transcription in the nucleus , translation in the cytoplasm	prokaryote: both in the cytoplasm, simultaneously
DNAthe template → mRNAthe message → PROTEINthe product TRANSCRIPTION in the nucleus, by RNA polymerase TRANSLATION at the ribosome; tRNA brings each amino acid The template is READ 3' to 5'. The new strand is BUILT 5' to 3'. Always. A codon is three bases on the mRNA; an anticodon is the matching three on the tRNA. The code is degenerate, so several codons give the same amino acid. Name the ENZYME and the LOCATION. Most explain marks want both.		

WORKED — READING A RIBOSOME FIGURE

Four amino acids from the anticodons shown 2

“State the amino acids, 1 to 4 on the growing amino acid chain in sequence. (2 marks)” The figure shows tRNA anticodons ACC, UUU, CUA and AAG against an mRNA reading UGGAAAGAUUUC.

- The letters on each tRNA are **ANTICODONS**. Looking those up directly gives the wrong four amino acids every time
- Read the mRNA instead, in threes: **UGG AAA GAU UUC**
- UGG → **Trp**, and the anticodon ACC confirms it
- AAA → **Lys** (anticodon UUU)
- GAU → **Asp** (anticodon CUA)
- UUC → **Phe** (anticodon AAG)

Trp – Lys – Asp – Phe, in that order

NITROGEN AND THE UREA CYCLE		
SIGNATURE	METHOD	EXAMPLE
the organ	the liver	not the kidney – the kidney only excretes urea
transamination	amino group moved to another keto acid	in the cytoplasm, nothing released
oxidative deamination	amino group released as free NH_3	in the mitochondria, and makes NADH
which compartment	ammonia, CO_2 , carbamoyl phosphate, citrulline formed → mitochondrion argininosuccinate, arginine, urea → cytoplasm	the cycle starts in the mitochondrion and finishes in the cytosol
the two nitrogens of urea	one from ammonia via carbamoyl phosphate one from aspartate	that framing makes the aspartate mark easy
the by-product when arginine forms	fumarate	never named in the deck, but labelled in the figure

NITROGEN AND THE UREA CYCLE		
SIGNATURE	METHOD	EXAMPLE
why convert ammonia at all	ammonia is a neurotoxin ; urea is neutral	raised blood ammonia causes hyperammonaemia and cerebral oedema
LIVER CELL cytoplasm TRANSAMINATION amino group moved to a keto acid nothing is released MITOCHONDRION OXIDATIVE DEAMINATION releases free NH_3, makes NADH $\text{NH}_3 + \text{CO}_2$ to carbamoyl phosphate, then to citrulline then in the CYTOPLASM citrulline to argininosuccinate to arginine to UREA The cycle STARTS in the mitochondrion and FINISHES in the cytoplasm. Ammonia is a neurotoxin; urea is neutral and safe to carry in blood.		

WORKED — WHICH PROTEIN AGGREGATES

Two partially folded proteins on heating 3

“When exposed to elevated temperature in vitro, state which Protein (Q or R) would form more protein aggregates. Explain your choice. (3 marks)” The figure shows identical folds; only the residue colouring differs — Q has hydrophobic residues buried, R has them on the surface.

- Use a printed word. The verdict is **Protein R**
- R has its **hydrophobic residues exposed on the surface**, in contact with water, whereas Q has them buried in its interior
- Heat disrupts the **weak non-covalent bonds** — hydrogen bonds, ionic interactions, Van der Waals — so the protein unfolds and exposes still more hydrophobic residues
- Exposed hydrophobic residues on **different molecules** then cluster together to minimise contact with water, and the molecules clump into aggregates
- Worth adding: the disulfide bonds are **covalent and heat-resistant**, so they are not what distinguishes the two

Protein R, because surface hydrophobic residues drive aggregation once heat unfolds the structure

ENZYMES, KINETICS AND INHIBITION		
SIGNATURE	METHOD	EXAMPLE
the one question that settles inhibition	does MORE SUBSTRATE rescue the rate?	yes → competitive no → non-competitive
competitive	Km rises, Vmax unchanged	binds the active site; usually a structural analogue of the substrate
non-competitive	Vmax falls, Km unchanged	binds an allosteric site and changes the enzyme shape
from a graph	same plateau, curve shifted right → competitive lower plateau, same half-point → non-competitive	
reversible or irreversible	is the bond covalent?	covalent → irreversible, as with penicillin, aspirin, orlistat
Km	the [S] that gives half Vmax	Low Km means HIGH affinity
reading Km off a figure	1. find the plateau, that is Vmax 2. halve it 3. read across to the curve, then down	state the unit – one part awards it explicitly
temperature or pH away from the optimum	weak non-covalent bonds break, the enzyme denatures	the active site loses its shape, so no enzyme-substrate complex forms

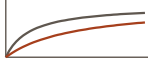
ENZYMES, KINETICS AND INHIBITION

SIGNATURE	METHOD	EXAMPLE
the six classes	oxidoreductase, transferase, hydrolase, lyase, isomerase, ligase	the deck writes the first as oxidase / dehydrogenase / reductase; the word oxidoreductase is in the notes only

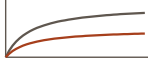
Does MORE SUBSTRATE rescue the rate?

YES ✓

NO ✗



COMPETITIVE
Km RISES, Vmax same
binds the ACTIVE site
looks like the substrate



NON-COMPETITIVE
Vmax FALLS, Km same
binds an ALLOSTERIC site
changes the enzyme shape

Km = the [S] that gives HALF Vmax.
Low Km means HIGH affinity.

The equation is not printed anywhere in the notes: $V = V_{max} [S] / (K_m + [S])$.

WORKED — INHIBITION FROM A FAMILY OF CURVES

Reading Vmax and Km off four curves

"What is the Vmax of the uninhibited enzyme? Please indicate the unit. What is the Km of enzyme when 1.0, 2.0 and 3.0 micromolar inhibitor are added? Is this inhibitor competitive or non-competitive? Explain the reasoning."

1. Uninhibited plateau = 40 mM/min.
The stem asks for the unit, so the unit is a mark

2. Half of 40 is 20; reading across and down gives Km = 40 mM

3. Repeat for each curve: plateaus fall to 20, 10 and 5 mM/min, but every curve reaches half its own plateau at 40 mM

4. So Vmax falls progressively while Km stays constant

5. Km unchanged means substrate affinity is unaffected, so the inhibitor is **not** at the active site — it binds elsewhere and distorts the enzyme

6. Therefore the inhibition **cannot be overcome by adding more substrate**

Non-competitive. Vmax 40 mM/min, Km 40 mM at every inhibitor concentration

WORKED — INHIBITION FROM A SENTENCE ALONE

Sulfanilamide and PABA

"When the concentration of the substrate PABA is increased, the inhibitory effect of sulfanilamide is reduced. Explain the possible mode of inhibition exhibited by sulfanilamide."

1. That sentence alone proves it.
Nothing else in the module behaves that way

2. Sulfanilamide is a **competitive inhibitor**

3. It is a **structural analogue** of PABA — both carry a benzene ring with a para-amino group — so its shape fits the same active site

4. The two therefore **compete for the same active site**, and only one can be bound at a time

5. Raising PABA raises the probability that PABA rather than the inhibitor occupies the site, so the inhibition is relieved

6. Kinetically: **Km rises, Vmax is unchanged**, because at high enough [S] the inhibitor is outcompeted

Competitive. The tell is in the stem, not the structures

TRANSPORT — COUNT THE ARROWS, THEN CHECK THE ENERGY		
SIGNATURE	METHOD	EXAMPLE
one mole- cule, one direction	uniporter	GLUT2
two differ- ent mole- cules, same direction	symporter	sym means same. SGLT1
two differ- ent mole- cules, op- posite directions	antiporter	anti means against. The sodium-potassium pump
channel or transporter	an open lined pore, many mole- cules at once → channel binds and changes shape → transporter	the words confor- mational change mean transporter
primary or secondary active transport	is ATP hy- drolysed in this step?	yes → primary. no, it rides an exist- ing gradient → secondary
the pump stoichiome- try	3 Na⁺ out, 2 K⁺ in , per ATP	
tonicity	water moves to the side with more solute	hypotonic outside → water in → the cell bursts hypertonic outside → water out → it shrinks

TRANSPORT — COUNT THE ARROWS, THEN CHECK THE ENERGY		
SIGNATURE	METHOD	EXAMPLE
which receptor	the tell is the LIGAND, not the receptor	hydrophilic ligand → cannot cross → cell surface hydrophobic, e.g. a steroid → dif- fuses through → intracellular
signalling mode	binds the cell that made it → autocrine contact only → juxtacrine neighbours → paracrine blood- stream → endocrine	the word blood- stream, or a named gland, settles it

WORKED — TWO QUESTIONS THAT BOTH ANSWER NO, FOR OPPOSITE REASONS

The receptor pair, which is a designed trap

"Would testosterone be able to activate estrogen receptor? **(3)**" and "Estrogen-Z is a hydrophilic molecule. Would Estrogen-Z be a suitable drug? **(3)**"

1. Both verdicts are **no**. **Memorising that loses half the marks** — the reasons are opposite

2. **Testosterone**: it is hydrophobic, so it **can** cross the membrane and reach the receptor. Delivery is not the problem

3. It fails on **binding**: ligand-receptor binding is specific in charge, shape,

size and hydrophobicity, and testosterone is a different shape, so it cannot induce the conformational change

4. **Estrogen-Z**: the stem states it **does** activate the receptor, so binding is not the problem

5. It fails on **delivery**: the receptor is **intracellular**, and a hydrophilic molecule cannot cross the hydrophobic core of the bilayer

6. Say **which gate fails** in each case

No and no. One fails at the door, the other at the lock

MODEL WORDING — TRANSPORT AND SIGNALLING

DO YOU AGREE THAT IT IS AN ANTIPORTER

Yes. An **antiporter** moves two different solutes in **opposite** directions. The pump moves **Na⁺** out of the cell and **K⁺** into it, which fits that definition.

WHY ENERGY IS NEEDED

Both ions are moved **against** their concentration gradients, from low to high concentration. That cannot happen spontaneously, so energy is required, supplied by **ATP hydrolysis**.

WHICH SIGNALLING MODE

Endocrine. The signal is released by **the ovary**, an endocrine gland, into the bloodstream, and travels a long distance to reach **distant target cells**, which is the defining feature of that mode.

REACTIVE OXYGEN SPECIES AND THE DEFENCES		
SIGNATURE	METHOD	EXAMPLE
oxidised or reduced	OIL RIG	oxidation is loss, reduction is gain, of electrons
the agents	an oxidising agent is it-self reduced	a reducing agent is itself oxidised. Antioxidants are reducing agents
free radical or just a ROS	does it have an UNPAIRED electron?	superoxide and hydroxyl yes. Hydrogen peroxide is a ROS but NOT a radical
reactivity order	hydroxyl far above superoxide, above nitric oxide	Lifetime under one second
where superoxide comes from	electron leak at Complex I and III	oxygen receiving an incomplete number of electrons
substrate is superoxide	SOD	two superoxide give one H ₂ O ₂ and one O ₂
substrate is H ₂ O ₂ , oxygen released	catalase , in the peroxisome	2H ₂ O ₂ → 2H ₂ O + O ₂
substrate is H ₂ O ₂ , glutathione mentioned	GPx	uses 2 GSH, which become GSSG
GSSG back to GSH	glutathione reductase , spending NADPH	NADPH comes from the oxidative steps of the PPP, rate-limited by G6PD

REACTIVE OXYGEN SPECIES AND THE DEFENCES

SIGNATURE	METHOD	EXAMPLE
the danger branch	H₂O₂ + Fe or Cu → the FENTON reaction	produces the hydroxyl radical, which damages DNA, protein and lipid
enzymatic or not	ends in -ase → enzymatic	made by the body → endogenous, e.g. glutathione. From the diet → exogenous, e.g. vitamin C

O₂ → leak at Complex I and III

↓

SUPEROXIDE O₂^{•−} a free radical

↓

I SOD

HYDROGEN PEROXIDE H₂O₂ NOT a radical

↙ ↘

CATALASE peroxisome 2H₂O + O₂

GPx uses 2 GSH 2H₂O

FENTON, Fe or Cu **HYDROXYL •OH** the most reactive

REGENERATION GSSG to 2 GSH, by glutathione reductase which spends NADPH, made by the PPP (G6PD)

Hydrogen peroxide is a ROS but NOT a free radical, having no unpaired electron. Reactivity: hydroxyl far above superoxide, above nitric oxide.

WORKED — A MUTATED CATALASE GENE

Consequence chain for a missing enzyme

"A gene that codes for catalase is mutated. Explain what would happen to the cell shown in Figure 2."

1. The mutation gives a non-functional catalase or none, so **H₂O₂ is no longer decomposed** to water and oxygen in the peroxisome

2. Hydrogen peroxide therefore **accumulates**. GPx compensates partly, but its capacity is exceeded because the three SOD isoforms keep producing more

3. Accumulated H₂O₂ reacts with iron or copper through the **Fenton reaction** to form the **hydroxyl radical**

4. The hydroxyl radical oxidises macromolecules non-specifically: DNA strand breaks and base mutation, enzyme inactivation, and lipid peroxidation that raises membrane fluidity and permeability

5. The result is oxidative stress, cellular dysfunction and potentially cell death

What builds up, what it turns into, and what that damages — all three

READING THE LUNG EPITHELIUM FIGURE

The ROS labelled are superoxide and hydrogen peroxide. Do not add the hydroxyl radical — it is not shown.

DUOX and NOX4 are not removal enzymes. They are NADPH oxidases that generate H₂O₂. On a question about enzymes that remove ROS they must be excluded, or better, named explicitly as producers to show the figure was actually read.

The three SOD isoforms cover three compartments: SOD3 in the lining fluid, SOD1 in the cytosol, SOD2 in the mitochondrion.

EXPLAINING A TWO-STAGE DEFENCE

Yes. **SOD** converts the more reactive **superoxide** into the less reactive **hydrogen peroxide**, and **catalase** then converts that into harmless water. The two act as a relay.

THE GAPS — examined but absent from every deck and note

EXAMINED, BUT ABSENT FROM EVERY DECK AND EVERY NOTE		
TOPIC	WHAT IS NEEDED	WHERE IT IS EXAMINED
gel electro-phoresis	DNA carries a net negative charge from the phosphate backbone, so it migrates towards the positive electrode (the anode)	a question asks which electrode is positive, from the direction of movement
agarose concentration	raising it from 1% to 5% gives a denser matrix with smaller pores, so all fragments move more slowly and bands compress towards the wells	small fragments resolve better; large ones may not separate at all
restriction enzymes	bind a specific short, usually palindromic recognition site, then hydrolyse the phosphodiester bond in each strand	a question asks how a restriction enzyme cuts DNA
cohesive against blunt	are the two cut positions offset or aligned?	offset → single-stranded overhangs → cohesive or sticky directly opposite → blunt

EXAMINED, BUT ABSENT FROM EVERY DECK AND EVERY NOTE		
TOPIC	WHAT IS NEEDED	WHERE IT IS EXAMINED
a restriction enzyme at 80 degrees	it is a protein: heat breaks the weak non-covalent bonds holding the tertiary structure, so it denatures and the active site no longer fits	answerable from the enzyme lesson, even though the context is not taught
uncouplers and UCP	ATP down, electron flow up, energy lost as heat	asked in full, and the word appears on no slide
lactate dehydro-genase	the enzyme of the muscle fermentation step	the paper anonymises it as Enzyme X; the name is nowhere in the module
mono-cistronic	one structural gene, one promoter, one protein	only polycistronic is ever defined, yet a question offers both
phospho-rylase kinase	it phosphorylates and so activates glycogen phosphory-lase	an entire question depends on it
protein phos-phatase-1	insulin activates it; it removes phosphate	appears as an option in two questions
Michaelis-Menten	$V = V_{\max} [S] \div (K_m + [S])$	the curve and both terms are taught; the equation is never printed

NUMBERS THE DECK LEAVES BLANK, AND WHERE THEY HIDE		
WHAT	VALUE	NOTE
the ATP convention	3 per NADH, 2 per FADH ₂ , 12 per acetyl-CoA, 38 and 36 per glucose	a textbook or a search gives 2.5, 1.5 and 30 to 32. That is a DIFFERENT CONVENTION, not a correction
the fasting reserves	carbohydrate about 1 day protein about 15 days fat about 85 days	at 1 600 kcal per day. The days column is blank in the deck
glycogen storage disorders	high glycogen, enlarged liver → breakdown blocked near-zero glycogen, normal liver → synthesis blocked	every disorder name lives in the speaker notes only
liver against muscle	only the liver has glucose-6-phosphatase	this distinction is in the notes pane, not on the slide
ketosis against ketoacidosis	ketones 0.5 to 3 mmol/L, pH normal → ketosis above 10 mmol/L, pH below 7.3 → ketoacido-sis	the tell is the pH number
two printing errors worth knowing	one slide titled G1 (Gap 2) describes G2 one slide headed Translation defines transcription	copying the first word off either slide loses a free mark